OPERATIONAL EXCELLENCE · WHAT THE RESEARCH SAYS

When there is no bed upstairs, patients wait in the emergency department. That wait has a cost.

Two recent studies from England and South Korea, covering millions of patients.

Swipe →

THE PROBLEM

How full beds turn into waiting

Emergencies don't arrive evenly. When the wards are full, a patient who needs admission has nowhere to go. They stay in the emergency department, and the department fills up with people waiting.

Bed occupancy is the share of available beds that are in use.

STUDY 1: ENGLAND

After five hours, every extra hour of waiting carried a cost

Researchers analyzed 5.2 million patients admitted from major emergency departments in England from 2016 to 2018. The study was published in 2022.

5 h

After five hours from arrival, 30-day mortality rose steadily the longer patients waited, up to 12 hours.

8%

The largest rise in mortality, seen in patients who waited more than six to eight hours.

WHAT THE NUMBERS MEAN

One extra death for every 82 patients

1 in 82

The study estimated one extra death for every 82 admitted patients whose move to a hospital bed was delayed beyond six to eight hours from arrival.

STUDY 2: SOUTH KOREA

100%

When emergency beds were full, the odds of death rose

Researchers analyzed about 26 million emergency visits from 2018 to 2022. When emergency department occupancy reached 100% or more, the odds of in-hospital death rose by about 12% in higher-level departments and 21% in lower-level ones.

BEYOND HOSPITALS

The same pattern shows up far from the hospital ward

Any system with uneven demand behaves in a similar way, whether it is a call center, a production line, or a team that handles requests. As it nears full capacity, queues grow much faster than the workload.

This point comes from queueing theory, the math of waiting lines. It is not a finding of the two studies.

WHAT I TAKE FROM THIS FOR LEADERS

Spare capacity is part of the design, not waste.

1

Track occupancy at peak times, not only the yearly average. The average hides the days when the system chokes.

2

Plan capacity for the swings in demand, not just the average.

3

Track how long patients wait for a bed. It shows how crowded the whole hospital is, not just the emergency department.

> If your team is busy all the time, where will tomorrow's urgent work go?

SOURCES AND LIMITS

What this evidence can and cannot tell us

Both studies are observational. They show association, not cause, although the researchers adjusted for age, other illnesses, and other factors.

A study in Stockholm found no link between hospital bed occupancy and mortality, so the effect may differ between health systems.

The studies come from England and South Korea, not from Arab health systems.

Editorial disclaimer: The views and analyses presented in this article represent the author's professional interpretation based on the cited sources and do not necessarily represent the official positions of the organizations mentioned. For the original, complete and most current information, please consult the primary sources and official documents listed in the references.

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